

FAISAL HASAN MAHIN

Address: 84/1 Kalabagan 1st Lane

Phone: +880 1984302439

Email: faisalsplaybook@gmail.com

LinkedIn: <http://www.linkedin.com/in/faisalsplaybook>

Github: <https://github.com/faisalsplaybook>

Portfolio: <https://faisalsplaybook.github.io/>



SUMMARY

A proactive final-semester Computer Science and Engineering student who has hands-on experience in building independent full stack flutter apps alongside an ongoing NLP research thesis detecting Bangla–English financial scam SMS, where a baseline model reached 95–98% accuracy across three datasets. Comfortable moving between state management, navigation, cloud backends, REST API integration, Clean Architecture and applied ML research in Python and scikit-learn.

TECHNICAL SKILLS

Languages & Frameworks: Dart, Flutter SDK, Object-Oriented Programming, Python

State Management: Riverpod (Provider, StreamProvider, FutureProvider, NotifierProvider), flutter_bloc (Cubit / BLoC)

Navigation: go_router, including StatefulShellRoute for tabbed navigation

Backend & Services: Firebase Authentication, Cloud Firestore, Firebase Cloud Messaging (FCM), flutter_local_notifications

Networking: REST API integration (repository-mediated), Dio with custom interceptors

Local Storage: Hive, shared_preferences

Architecture: Clean Architecture, feature-first (domain / data / presentation), Repository Pattern

Machine Learning: pandas, NumPy, scikit-learn, TF-IDF vectorization, Logistic Regression, Google Colab

Tools & Practices: Git, GitHub, flutter_dotenv, Unit Testing

AI Models & Agents: ChatGPT(codex), Claude(claude code & design), Gemini, Antigravity IDE etc.

Languages: Bangla — Native | English (Fluent, professional working proficiency)

PROJECTS

KiDekhba — Collaborative Movie & TV Watchlist App

- Made watching movies/tv series fun and interactive by building a Flutter app to jointly create, vote, and manage watchlists
- Integrated Firebase Auth, Firestore, and FCM for real-time sync and push notifications, verified across all three delivery states: foreground, background, terminated
- Applied Clean Architecture (feature-first domain/data/presentation, repository pattern); used Riverpod for app-wide state and Cubit/BLoC for search
- Integrated the TMDB REST API via Dio, using custom interceptors for authentication, error handling, and logging
- Implemented tabbed navigation with GoRouter's StatefulShellRoute across four sections, with unread-alert badges and invite accept/decline flows

SplitFrens — Bill-Splitting App

- Built a Flutter app for splitting shared group expenses, using Riverpod for state management and GoRouter for navigation
- Integrated a live currency-exchange-rate REST API via Dio for real-time multi-currency splitting, with Hive-backed offline persistence for bills and groups

RESEARCH PROJECT

Robust Detection of Mobile Financial Scams in Bangla-English SMS

Final Year Design Project, Daffodil International University — Ongoing (Phase-I complete)

- Standardized three SMS datasets: BanglaBarta, a Bangla spam corpus, and the UCI SMS Spam Collection (~11,000 messages) into a unified scam/non-scam scheme, with preprocessing for Bangla-English (Banglish) code-mixed text
- Trained and evaluated baseline TF-IDF + Logistic Regression classifiers, reaching up to 97.85% accuracy, F1 0.91-0.95 across all three datasets
- Extending to transformer fine-tuning (BanglaBERT / XLM-R) and robustness testing against paraphrased, obfuscated scam variants for Phase-II

EDUCATION

Daffodil International University, Dhaka

B.Sc. in Computer Science and Engineering | CGPA: 3.45 / 4.00

Jul 2021 – Sep 2026 (Expected)